Java Classes and Objects

Learning Objectives

By the end of the lesson, students should be able to:

- Define a **class** and an **object** in Java
- Explain the relationship between classes and objects
- Create classes with attributes and methods
- Instantiate objects and access their members
- Understand constructors and basic encapsulation

Lesson Breakdown

1. Introduction (10 minutes)

Goal: Build intuition

- Ask: *"What is a real-world example of a blueprint?"*
- Explain:
    - A **class** = blueprint/template
    - An **object** = real-world instance of that blueprint

Example analogy:

- Class: Car
- Objects: Toyota, BMW, Tesla cars

2. What is a Class? (15 minutes)

Definition:
A class is a user-defined data type that contains:

- Variables (attributes)
- Methods (functions)

Syntax:

```
class ClassName
{
   // attributes
   // methods
}
```

Example:

```
class Student
{
   String name;
   int age;
   void displayInfo()
{
      System.out.println(name + " is " + age + " years old.");
 }
}
```

Key Points:

- Classes are templates
- No memory allocated until objects are created

3. What is an Object? (15 minutes)

Definition:
An object is an instance of a class.

Creating an Object:

```
Student s1 = new Student();
```

Accessing Members:

```
s1.name = "Alice";

s1.age = 20;

s1.displayInfo();
```

Explain:

- new keyword allocates memory
- Dot (.) operator accesses properties

4. Constructors (10–15 minutes)

Definition:
A constructor is a special method used to initialize objects.

Example:

```
class Student

{

   String name;

   int age;


   Student(String n, int a)

  {

      name = n;

      age = a;

   }

}
```

Creating object with constructor:

```
Student s1 = new Student("Alice", 20);
```

Key Points:

- Same name as class

- No return type
- Automatically called

5. Encapsulation Basics (10 minutes)

Concept:

- Protect data using **private variables**
- Access via **getter/setter methods**

Example:

```
class Student {
   private String name;

   public void setName(String n) {
      name = n;
   }

   public String getName() {
      return name;
   }
}
```

6. Guided Practice (10–15 minutes)

Activity:
Students create a class Car with:

- Attributes: brand, speed
- Method: display details
- Constructor to initialize values

Expected Output Example:

```
Car c1 = new Car("Toyota", 120);

c1.display();
```

7. Common Mistakes to Address (5 minutes)

- Forgetting new keyword
- Confusing class vs object
- Missing semicolons
- Not using correct method calls

8. Summary (5 minutes)

Recap:

- Class = blueprint
- Object = instance
- Constructors initialize objects
- Encapsulation protects data

Assessment / Homework

Short Questions:

1. Define a class and an object
2. Write a Java class Book with attributes and a method

Programming Task:

- Create a BankAccount class with:
    - balance
    - deposit() and withdraw() methods